

PANKAJ KUMAR PRAMANIK

Senior Data & AI Engineer

Bangladesh | +880 1716 121009 | pkp2.me2k9@gmail.com | LinkedIn | GitHub | Website

PROFESSIONAL SUMMARY

Senior Data & AI Engineer with 8+ years of engineering experience spanning full-stack development, data engineering, analytics engineering, generative AI, agentic AI, MLOps/LLMOps, and automation. Builds production-grade multi-agent LLM systems, RAG applications, ETL/ELT pipelines, ML workflows, and AI voice/CRM automations integrated with real business processes. Delivers end to end across data ingestion, modeling, APIs, CI/CD, deployment, guardrails, observability, and monitoring on AWS, GCP, Azure, and VPS infrastructure. Open to international remote roles and contract engagements.

CORE SKILLS

- **Generative & Agentic AI:** LLMs, RAG, AI Agents, Multi-Agent Systems, LangGraph, LangChain, CrewAI, LlamaIndex, Model Context Protocol (MCP), OpenAI, Claude, Pinecone, FAISS, Hugging Face, Hermes Agent, Prompt Engineering, Tool Calling, Human-in-the-Loop, LLM Evaluation, AI Guardrails
- **LLMOps & Observability:** LangSmith, LangGraph Checkpointing, Structured Logging, Automated Testing, Monitoring, Multilingual Support
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, NLP, Transformers, CNNs, Transfer Learning, Grad-CAM, Recommender Systems, Predictive Modeling
- **Data Engineering & Analytics:** Python, SQL, ETL/ELT, Mage.ai, Airflow, dbt, Spark, Kafka, Databricks, Snowflake, BigQuery, Redshift, PostgreSQL, Star Schema Modeling, Pandas, Looker Studio, Power BI, Tableau
- **MLOps, Cloud & DevOps:** MLflow, DVC, Kubeflow, Docker, Kubernetes, GitHub Actions, GHCR, CI/CD, AWS (EC2, ECR, SageMaker, Lambda, S3, RDS), GCP, Azure, OVHcloud, VPS Deployment
- **Automation & Full Stack:** n8n, Make.com, Retell AI, GoHighLevel, Twilio, FastAPI, Flask, Django, Streamlit, React, TypeScript, Next.js, Node.js, Supabase, SQLAlchemy, Stripe, Zoho CRM, Resend, REST APIs, Webhooks

PROFESSIONAL EXPERIENCE

AI Engineer

Mar 2025 - Present

Robinson & Associates, Maryland, USA

Remote

- Built automated lead intake, qualification, routing, follow-up, and case-status workflows using n8n, Make.com, Google Sheets, and APIs.
- Developed AI outbound calling with Retell AI covering lead qualification, scheduled callbacks, live-agent transfers, voicemail handling, and call-result tracking.
- Integrated OpenAI and Perplexity for legal FAQ support, content enrichment, and AI-assisted workflow decisions.
- Created GoHighLevel workflows for lead tagging, routing, appointment scheduling, communication tracking, DND/opt-out management, and pipeline updates.
- Built an n8n error-monitoring workflow that captures failed executions and sends them to an AI debugging agent for diagnosis.
- Implemented retry controls, STOP/DND handling, validation rules, and scheduling safeguards, and unified transcripts, SMS replies, transfers, and case statuses into structured reporting.

Generative AI, Data Engineering & MLOps Specialist

Jan 2025 - Present

Upwork (Freelance / Contract)

Remote

- Build RAG applications with LangChain, GPT-4, embeddings, and Pinecone/FAISS retrieval pipelines for knowledge-grounded answers.
- Develop AI agents, voice agents, chat assistants, and workflow automations using n8n, Retell AI, APIs, webhooks, and Google Workspace.
- Build ETL/ELT pipelines with Airflow, dbt, Mage.ai, Pandas, SQL, and BigQuery, and deliver dashboards in Looker Studio, Power BI, and Tableau.
- Apply MLflow, DVC, Kubeflow, and AWS SageMaker for experiment tracking, model versioning, deployment, and reproducible ML workflows.

AI SaaS Developer

May 2026 - Jul 2026

Seiton Paw, Florida, USA

Contract / Remote

- Built an AI pet-travel compliance platform (React, FastAPI, Supabase/PostgreSQL) in which a LangGraph agent reads country-specific destination rules and generates required tasks, deadlines, documents, and a readiness score.
- Integrated Stripe checkout and payment reconciliation, Zoho CRM sync, Resend transactional email, scheduled reminders, and full EN/ES localization.
- Delivered trip management, billing, document upload, and validation workflows, replacing manual research with a guided flow that catches errors before departure.

Generative AI & Full-Stack Developer

May 2024 - Jan 2025

SubHub, New York, USA

Remote

- Developed RAG/NLP workflows using LangChain, Pinecone, and LangFlow with semantic retrieval to improve contextual search and AI responses.

- Built real-time features with Supabase Realtime/PostgreSQL and scalable Next.js and Python services.
- Integrated Twilio SMS and ElevenLabs voice for automated and conversational customer communication.

Machine Learning Engineer

Most Loved Workplace, USA

May 2023 - Dec 2023

Remote

- Developed statistical ML, text analytics, data-mining, and predictive models on large structured and unstructured datasets.
- Translated business requirements into analytical models and communicated findings with Python, Pandas, NumPy, Scikit-learn, and Matplotlib.

SELECTED PROJECTS

TravelCrewAI - Multi-Agent LLMops Travel Planning Platform | IIT Guwahati

Jul 2026 - Sep 2026

Production-oriented multi-agent travel planner where a LangGraph supervisor routes requests to Flight, Hotel, Weather, Budget, and Itinerary agents, with human-in-the-loop approval, selective agent re-execution, checkpointing, and guardrails. External services connect through MCP, and PostgreSQL stores trips, conversations, reviews, and audit data. Includes structured logging, LangSmith observability, multilingual support, automated tests, and CI/CD with Docker, GitHub Actions, GHCR, and an OVHcloud VPS. Stack: Python, FastAPI, LangGraph, LangChain, MCP, React, TypeScript, PostgreSQL, SQLAlchemy.

Uber Data Analytics - Modern Data Engineering on GCP

Apr 2025 - Aug 2026

End-to-end ETL on NYC TLC trip data: Mage.ai extraction, validation, and transformation into a BigQuery star schema, a flat analytics table for Looker Studio dashboards, and a dbt layer adding lineage, column-level tests, and derived metrics.

U.S. Visa Approval - End-to-End MLOps | IIT Guwahati

Sep 2025 - Jan 2026

Classification pipeline from data ingestion to monitoring with MLflow tracking, Docker, AWS ECR/EC2, and GitHub Actions CI/CD, reaching 87% accuracy.

Bird Disease Classification - Deep Learning & MLOps | IIT Guwahati

Feb 2026 - May 2026

Healthy vs. coccidiosis classification from poultry fecal images using transfer learning (VGG16, ResNet50, EfficientNetB0, MobileNetV2) with MLflow/DVC, Grad-CAM, FastAPI, and Docker on AWS.

CareSage - Medical RAG Assistant

Jan 2026 - Apr 2026

PDF ingestion, chunking, embeddings, Pinecone vector search, and a LangChain retrieval pipeline producing grounded answers, served through a Flask backend and chat UI.

MovieMate - Personalized Movie Recommender

May 2026 - Jul 2026

Content-based recommender using cosine similarity, with a Streamlit interface and TMDB API integration returning similar movies with posters and metadata.

SELECTED ACHIEVEMENTS

- **Multi-Agent Production Delivery:** Designed and deployed TravelCrewAI, a supervisor-based multi-agent LLM platform, to a production VPS with human-in-the-loop approval, guardrails, LangSmith tracing, and CI/CD.
- **Agentic Workflow Automation:** Replaced manual pet-travel compliance research at Seiton Paw with a LangGraph agent that generates country-specific checklists, deadlines, and readiness scores, integrated with Stripe, Zoho CRM, and EN/ES localization.
- **Reliable AI Operations:** Automated legal lead intake, AI calling, SMS follow-up, scheduling, and CRM updates at Robinson & Associates, with retry, STOP/DND, and duplicate-prevention safeguards and an AI-assisted n8n error-diagnosis workflow.
- **Analytics Data Quality:** Delivered a GCP ELT pipeline into a BigQuery star schema with a dbt layer adding lineage and column-level tests for trusted Looker Studio reporting.
- **End-to-End MLOps:** Reached 87% accuracy on a U.S. visa approval classifier with MLflow-tracked experiments and automated deployment to AWS ECR/EC2 via GitHub Actions.

EDUCATION

B.Sc. in Data Science and Artificial Intelligence

Indian Institute of Technology Guwahati

Sep 2023 - Oct 2027 (Expected)

India

B.Sc. in Mechanical Engineering

Khulna University of Engineering & Technology

2009 - 2015

Bangladesh

CERTIFICATIONS

- **Generative AI Essentials: A Comprehensive Introduction** - IBM (Coursera)
- **Developing AI Applications with Python and Flask** - IBM (Coursera)
- **Introduction to DevOps** - IBM (Coursera)
- **Data Science Certificate Program** - Ostad
- **The Complete Guide to Becoming a Software Architect** - Udemy
- **Introduction to Software Engineering** - IBM (Coursera)